

BioCoLoop: PUBLIC-HARNESS COMPARISON

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Public frameworks use their task-adapted native experimental controllers and the common biological fitting interface. AI-Scientist-v2 and AI-Researcher access laboratory 0, whereas BioCoLoop accesses ten laboratories. Matched Qwen2.5-7B-Instruct and GPT-5.6 Luna blocks use the same twelve-design library, at most six candidate proposals and the same held-out scorer. Scored cells pass independent recomputation from saved predictions; failure tokens record terminal controller outcomes rather than predictive scores.

Model / method	DTI	Proteomics	Cell perturbation			Average
	BindingDB AUROC $\uparrow$	PTPC AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$	Mean rank $\downarrow$
Qwen2.5-7B-Instruct						
Fixed model (1 lab)	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00	2.40
Direct (1 lab)	<u>89.89</u> $\pm$ 1.02	<u>20.13</u> $\pm$ 0.07	<u>21.09</u> $\pm$ 4.38	14.15 $\pm$ 0.00	<u>15.42</u> $\pm$ 6.21	3.40 $\downarrow$ 1.0
AI-Scientist-v2 (1 lab)	89.68 $\pm$ 0.80	20.18 $\pm$ 0.43	18.61 $\pm$ 2.78	14.12 $\pm$ 0.04	15.42 $\pm$ 6.21	3.80 $\downarrow$ 1.4
AI-Researcher (1 lab)	86.09 $\pm$ 0.88	20.12 $\pm$ 0.05	18.61 $\pm$ 2.78	21.73 $\pm$ 6.63	17.41 $\pm$ 6.21	3.80 $\downarrow$ 1.4
BioCoLoop (10 labs)	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59	1.20 $\uparrow$ 1.2
GPT-5.6 Luna (low reasoning)						
Fixed model (1 lab)	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00	2.20
Direct (1 lab)	<u>89.89</u> $\pm$ 1.02	<u>20.17</u> $\pm$ 0.42	<u>18.61</u> $\pm$ 2.78	21.75 $\pm$ 6.59	<u>15.92</u> $\pm$ 6.03	3.00 $\downarrow$ 0.8
AI-Scientist-v2 (1 lab)	89.59 $\pm$ 0.95	23.37 $\pm$ 5.89	18.61 $\pm$ 2.78	14.12 $\pm$ 0.04	11.94 $\pm$ 2.11 [†]	3.25 $\downarrow$ 1.0
AI-Researcher (1 lab)	–	–	–	–	–	–
BioCoLoop (10 labs)	93.67 $\pm$ 0.98	37.11 $\pm$ 0.01	31.50 $\pm$ 0.00	<u>22.69</u> $\pm$ 0.73	22.39 $\pm$ 2.59	1.20 $\uparrow$ 1.0

Table 1: Main comparison by research model (mean $\pm$ sample SD, seeds 42–44; scores $\times 100$). Public controllers use laboratory 0 and BioCoLoop ten under a shared design library, fits and held-out scorer. Bold/underline mark the best/second-best complete result; $\uparrow$ partial and – unscored cells are not imputed; the last column averages each method’s rank over completed endpoints (Appendix C).

Outcome. Each model block reports the three-seed mean and sample SD for every complete cell, marks a partial cell with a dagger and an unscored cell with a dash; Section C.2 of the manuscript lists the completed count and typed failure of every incomplete cell. Bold and underline identify the best and second-best complete values within each model and endpoint; no incomplete controller cell is ranked. The final column gives the mean rank over each method’s complete endpoints and its colored change against the block’s fixed reference.